ABSTRACT

Food Ingredient Recognition and Recipe Recommendation System Using DL

With the increasing demand for intelligent food recommendation systems, automatic ingredient recognition from food images has gained significant research interest. This project presents a cloud-based food ingredient recognition and recipe recommendation system that integrates deep learning, content-based filtering, and cloud computing technologies. The system automatically identifies fruits and vegetables from user-uploaded images using a Convolutional Neural Network (CNN) based on transfer learning with a pretrained ResNet50 model, trained on a publicly available fruits and vegetables image dataset.

After ingredient recognition, the system performs recipe recommendation using a content-based filtering approach on the Food.com recipe dataset. Recipe ingredient lists are processed using TF-IDF vectorization, and cosine similarity is applied to match the predicted ingredient with relevant recipes. Additionally, nutrition-aware filtering, such as calorie-based constraints, is supported to enhance personalized and health-conscious recommendations. To improve coverage and provide real-time results, the system optionally integrates the Edamam Recipe API as a supplementary source.

The application is developed using the Flask web framework and deployed on the Amazon Web Services (AWS) platform. Amazon EC2 is used for hosting the web application, Amazon S3 for storing uploaded images and trained model files, and AWS IAM for secure access management. This project demonstrates the effective integration of computer vision, text-based recommendation techniques, external APIs, and cloud deployment to build a scalable and practical food recommendation system for real-world applications.

Dataset:https://www.kaggle.com/datasets/kritikseth/fruit-and-vegetable-image-recognition
https://www.kaggle.com/datasets/irkaal/foodcom-recipes-and-reviews

Technology Stack:

- **Language:** Python
- **Machine Learning Libraries:** PyTorch, Torchvision, Pandas, NumPy
- **Web App Framework:** Flask
- **Deployment:** AWS (EC2, S3)
- **IDE & Tools:** VS Code, Google Colab

References:

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